

NPTEL Course on "Computer System Optimization for ML"

Lecture #39

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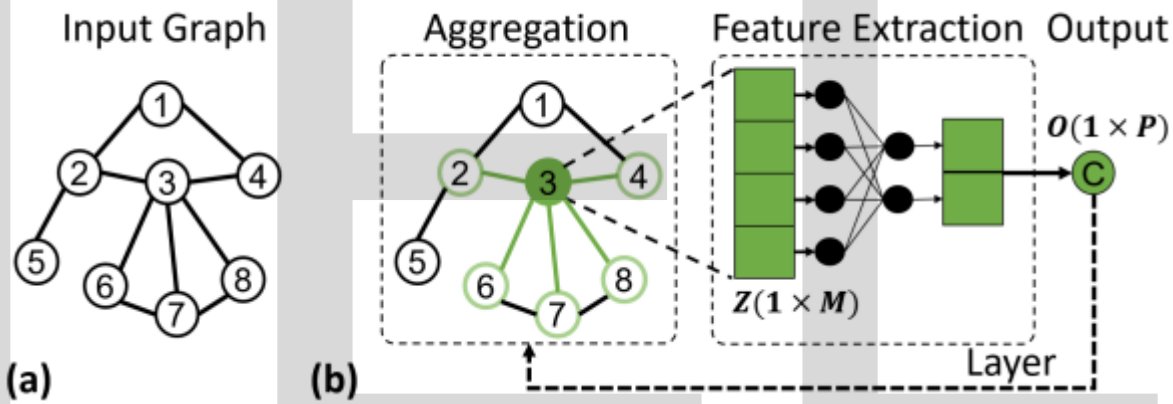
Agenda

- Recap
- GNN Accelerator

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Classification of Graph Nodes through GNN

- **Two phases**
  - Aggregation
  - Combination/Feature Extraction
- **Aggregate function aggregates multiple feature vectors from neighbours to form one single feature vector**
  - Different GNNs use different function
- **Combination function transforms the feature vector of each vertex to another feature vector using a MLP**
- **MLP parameters are shared between vertices**



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Summary of GNN Computations

- Several operations
  - Aggregation, Combination, Pooling, Readout
- Except combination, all the operations are graph structure-dependent, which involve graph processing
- Combination is a typical MLP
- Sampling is to sample a subset from neighbours
- Aggregation aggregates the features from one hop neighbours
- Pooling acts like a pooling in CNN
- Readout can be a simple summation across vertices; can be seen as extreme aggregation
- Aggregation and Combination are the two major phases

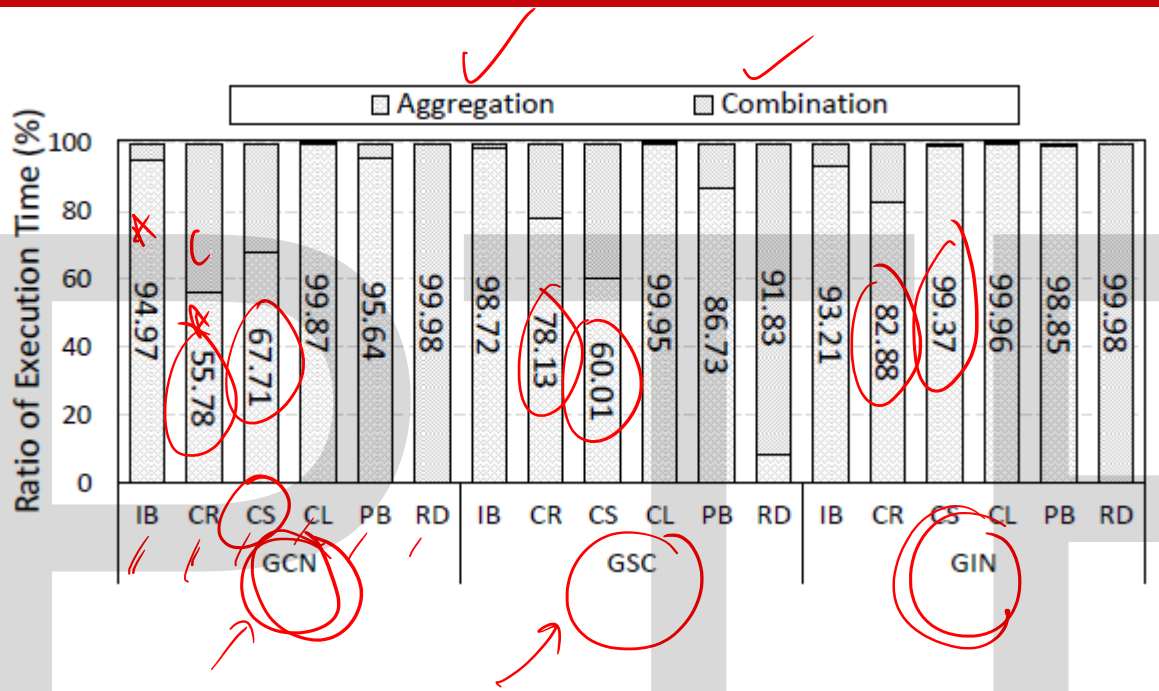
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Characterization of CPU



Dataset	#Vertex	Feature Length	#Edge	Storage
IMDB-BIN (IB)	2,647	136	28,624	1.5MB
Cora (CR)	2,708	1,433	10,556	15MB
Citeseer (CS)	3,327	3,703	9,104	47MB
COLLAB (CL)	12,087	492	1,446,010	28MB
Pubmed (PB)	19,717	500	88,648	38MB
Reddit (RD)	232,965	602	114,615,892	972MB

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Hybrid Execution Pattern

- Aggregation Phase heavily relied on the graph structure
  - Random and Sparse
  - Results in irregular access patterns
  - More data to be accessed from DRAM than combination
  - More misses per kilo instructions (MPKI)
- Combination performs matrix vector multiplication
  - Does not require much data
  - 36% of the time is spent on data copy

QUANTITATIVE CHARACTERIZATION ON CPU.

	Aggregation	*Combination*
DRAM Byte per Ops	11.6	0.06
DRAM Access Energy per Ops	$170nJ$	$0.5nJ$
L2 Cache MPKI	11	1.5
L3 Cache MPKI	10	0.9
Ratio of Synchronization Time	—	36%

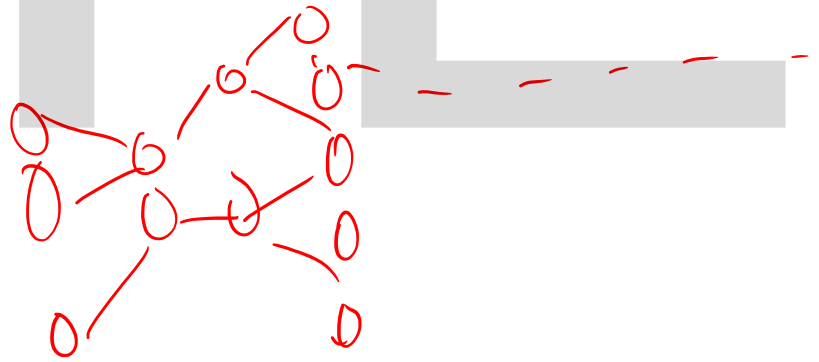
DIFFERENT EXECUTION PATTERNS OF *Aggregation* PHASE AND *Combination* PHASE.

	Aggregation	*Combination*
Access Pattern	Indirect & Irregular	Direct & Regular
Data Reusability	Low	High
Computation Pattern	Dynamic & Irregular	Static & Regular
Computation Intensity	Low	High
Execution Bound	Memory	Compute

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Differences from Conventional Workloads

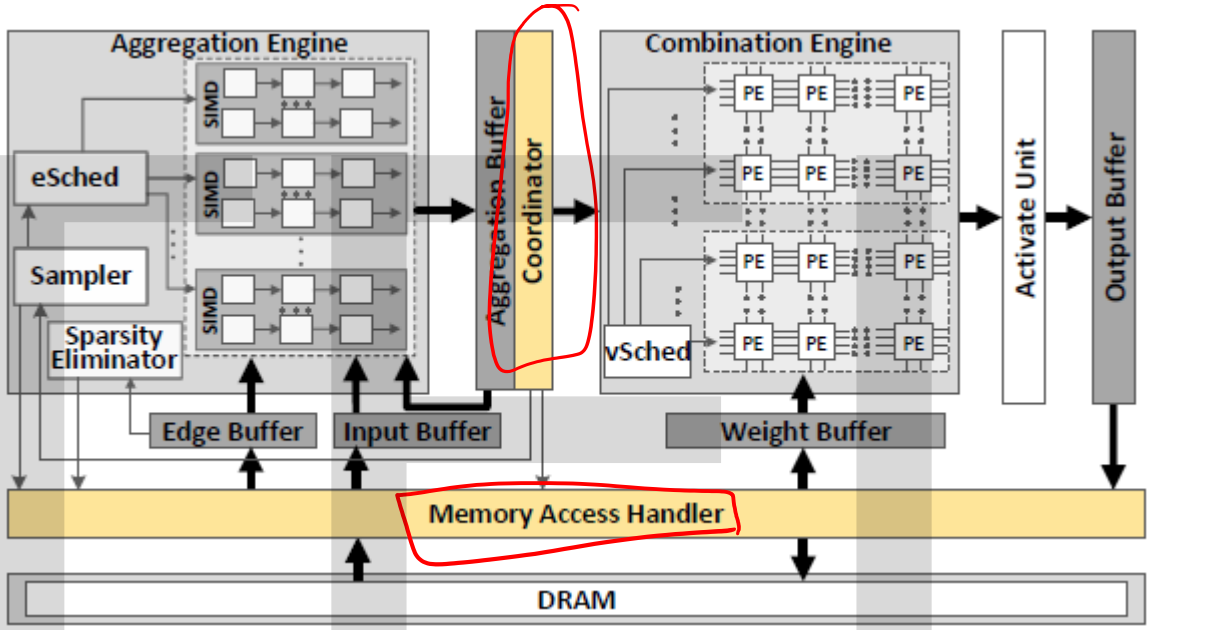
- In the Aggregation phase, the length of feature vector is variable rather than fixed
  - Determined by the MLP model
- In the Combination phase, the MLP parameters are fully shared by all vertices
  - Highly reusable inter-vertex data
- Two phases are executed alternatively



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HyGCN Architecture [1]

- HyGCN
  - A hybrid architecture
  - Two engines: Aggregation and Combination
  - Memory access Handler
  - A communication interface (coordinator)

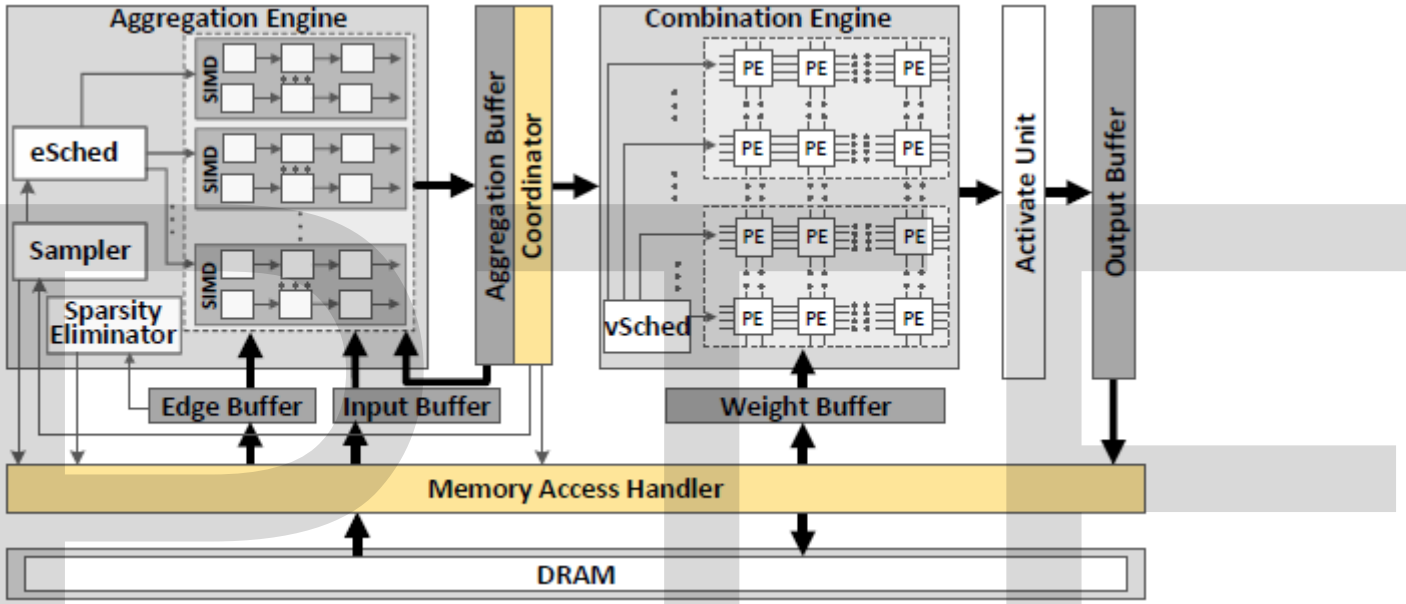




[1] Yan, Mingyu, et al. "Hygcn: A gcn accelerator with hybrid architecture." 2020 IEEE International Symposium on High Performance Computer Architecture (HPCA). IEEE, 2020.

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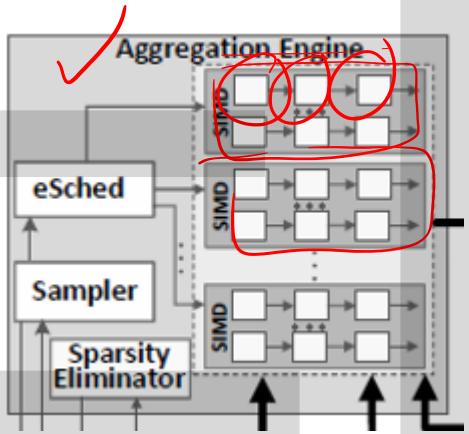
HyGCN Architecture



- Aggregation engine aims to realize the efficient execution of irregular accesses and computations
- Combination engine is designed to maximize the efficiency of regular accesses and computations
  - Systolic array based design

Aggregation Engine

- Static graph partition to enhance data reuse
- Dynamic sparsity elimination technique to reduce unnecessary data access
- Two execution mode
  - Vertex concentrated
  - Vertex disperse
- Vertex Concentrated mode
  - Each vertex is assigned to a single SIMD core
  - Can produce the aggregated features in burst mode
  - Processing latency of a single vertex can be long
    - Fast vertices need to wait for the slow vertices
    - Less parallelism



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Aggregation Engine

- Vertex Disperse Mode
- Aggregation of elements inside the vertex feature vector of each vertex to all cores
- If a vertex cannot occupy all cores, free cores can be assigned to other vertices
- All cores are always busy without workload imbalance

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THANK YOU

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